

PHASE 3

FitBuddy- AI Fitness Plan Generator using Gemini Model

Project Design

1. Introduction

The Project Design Phase is an important stage in the development of the **FitBuddy – AI Fitness Plan Generator**. After identifying the project idea during the Brainstorming & Ideation Phase and defining the system requirements during the Requirement Analysis Phase, the next step is to design how the complete system will work.

The main purpose of this phase is to transform the identified requirements into a structured technical design. The design defines how the user interface, FastAPI backend, Generative AI model, database, and application components communicate with each other.

FitBuddy is designed to generate personalized fitness plans based on information provided by the user. The documented project uses **FastAPI, Google Gemini, SQLite with SQLAlchemy, Jinja2, HTML/CSS, and Uvicorn** as its main technologies.

The design also considers the feedback process, where users can provide feedback on a generated workout plan and the system can generate an updated plan.

2. Purpose of Project Design

The main purposes of the Project Design Phase are:

- To define the overall architecture of FitBuddy.
- To identify the major components of the system.
- To define communication between the frontend and backend.
- To design the interaction between FastAPI and the Generative AI model.
- To design the database structure.
- To define the user workflow.
- To design the feedback and workout-plan revision process.
- To organize the application into manageable components.
- To provide a foundation for the development phase.

A good system design helps ensure that the different parts of FitBuddy can work together in an organized manner.

3. FitBuddy System Design Overview

The FitBuddy system is designed around a user-centered workflow. The user provides information such as:

- Name/User ID
- Age
- Weight
- Fitness goal
- Workout intensity

This information is processed by the backend and passed to the AI-based workout generation functionality. The generated workout plan is then displayed to the user through the frontend.

The system also provides a nutrition/recovery tip and allows the user to submit feedback. Based on feedback, the system can generate an updated workout plan.

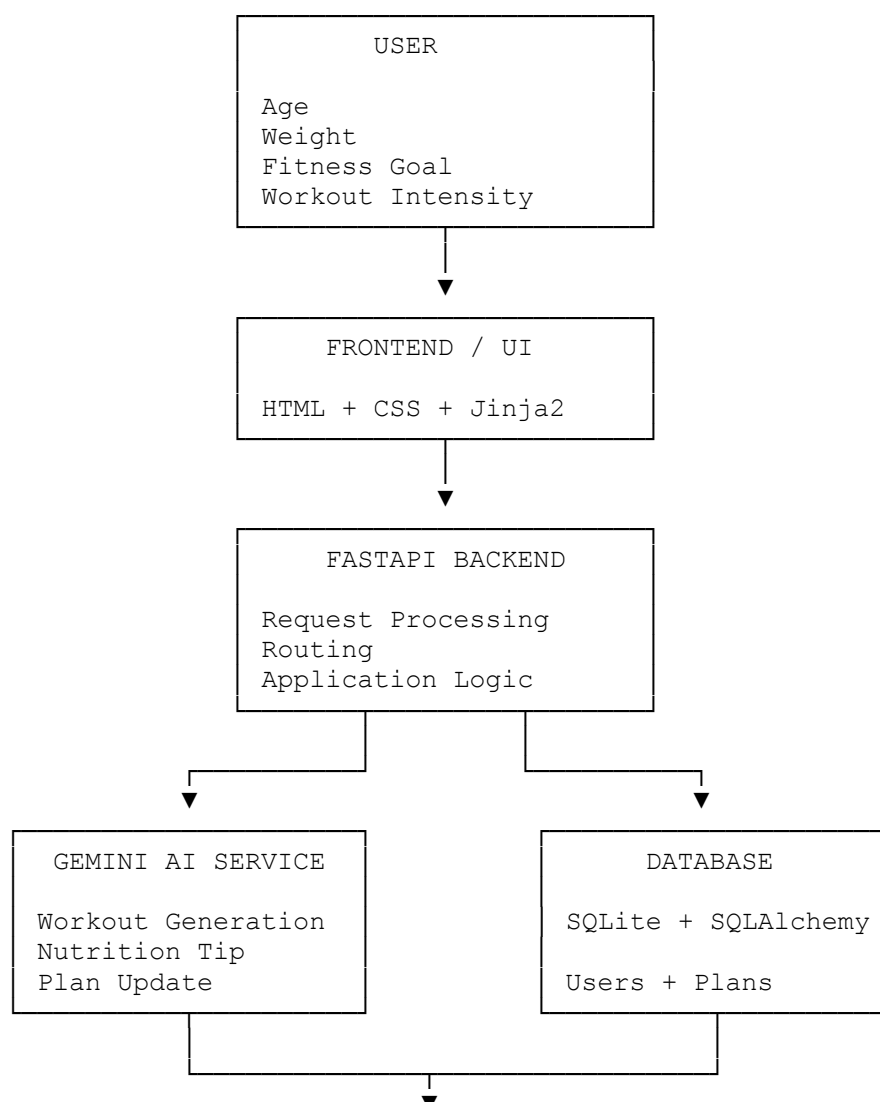
The major design components are:

1. User Interface
2. FastAPI Backend
3. AI/Generative AI Service
4. Database
5. Feedback Processing
6. Updated Workout Plan
7. User/Plan Viewing Interface

4. High-Level System Architecture

The high-level architecture describes how the major components of FitBuddy interact.

Figure 1: High-Level FitBuddy Architecture



GENERATED RESULT

7-Day Workout Plan
Nutrition Tip
Updated Plan

The architecture separates the user interface, backend processing, AI functionality, and data storage into logical components.

5. Architectural Layers

The FitBuddy design can be understood through several logical layers.

5.1 Presentation Layer

The presentation layer is responsible for interacting with the user.

The documented project uses:

- HTML
- CSS
- Jinja2 templates

The interface collects user information and displays generated workout plans and other results.

The source project includes pages such as:

- `index.html`
- `result.html`
- `all_users.html`

5.2 Application Layer

The application layer is implemented using **FastAPI**.

It manages:

- User requests
- Routing
- Input processing
- Calling AI functions
- Database operations
- Returning results to templates

The documented routes include:

- `/generate-workout`

- /submit-feedback
- /view-all-users

5.3 AI Processing Layer

The AI processing layer provides the Generative AI functionality.

The documented project uses Google Gemini for:

- Workout generation
- Nutrition/recovery tip generation
- Workout plan modification

The AI service receives structured information from the backend and generates text-based fitness recommendations.

5.4 Data Layer

The data layer is responsible for storing information.

The documented system uses:

- SQLite
- SQLAlchemy

The database stores user information and workout plans, including the original and updated plans.

6. Component Design

The major FitBuddy components are designed as follows.

Component 1 – User Interface

Responsible for:

- Collecting user information.
- Displaying workout plans.
- Displaying nutrition/recovery information.
- Collecting feedback.
- Providing access to user and plan information.

Component 2 – FastAPI Application

Responsible for:

- Receiving requests.
- Processing input.

- Managing routes.
- Calling AI functionality.
- Interacting with the database.
- Returning responses.

Component 3 – AI Service

Responsible for:

- Generating workout plans.
- Generating nutrition/recovery tips.
- Updating workout plans based on feedback.

Component 4 – Database

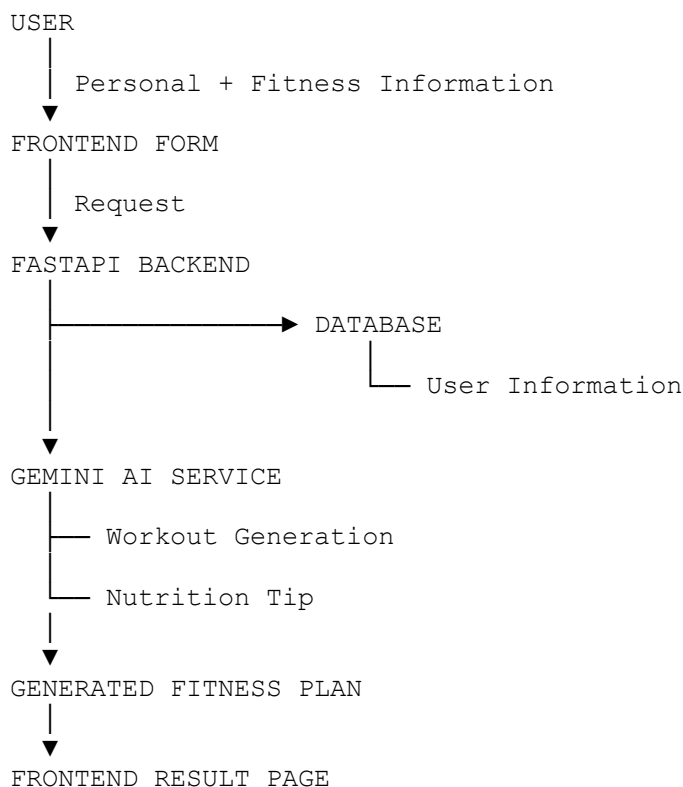
Responsible for:

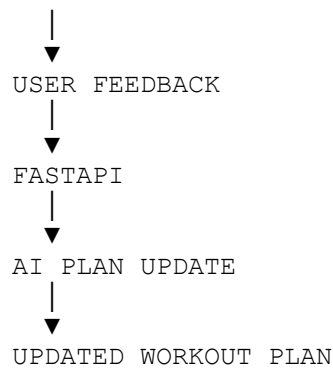
- Saving users.
- Saving generated plans.
- Updating plans.
- Retrieving users and plans.

7. Data Flow Design

The data flow describes how information moves through the FitBuddy application.

Figure 2: FitBuddy Data Flow





This design creates a continuous interaction between the user, backend, AI service, and database.

8. Database Design

The FitBuddy project uses **SQLite with SQLAlchemy** for data persistence.

The database design supports storing information about users and their generated plans.

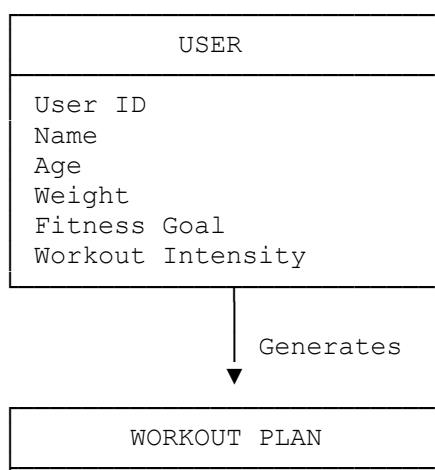
The documented database operations include:

- `save_user()`
- `save_plan()`
- `update_plan()`
- `get_original_plan()`
- `get_user()`
- `get_all_users()`
- `get_all_plans()`

These operations allow the application to create, retrieve, and update information.

9. Database Relationship Concept

Figure 3: Database Design Concept



Original Plan
Updated Plan
Feedback

The design allows user information to be associated with generated fitness plans.

The source document specifically describes persistence of the original and updated workout plans.

10. AI Integration Design

Generative AI is one of the central components of FitBuddy.

The documented implementation uses Google Gemini.

The AI functionality is separated into specific operations.

Main AI functions

10.1 `generate_workout_gemini()`

This function is responsible for generating the personalized workout plan.

It uses user-provided fitness information to generate the workout content.

10.2 `generate_nutrition_tip_with_flash()`

This function generates a concise nutrition or recovery-related tip.

The source project documents a Gemini Flash model for this functionality.

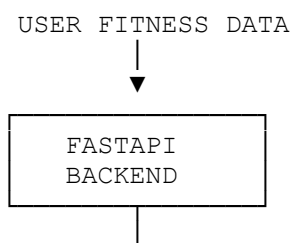
10.3 `update_workout_plan()`

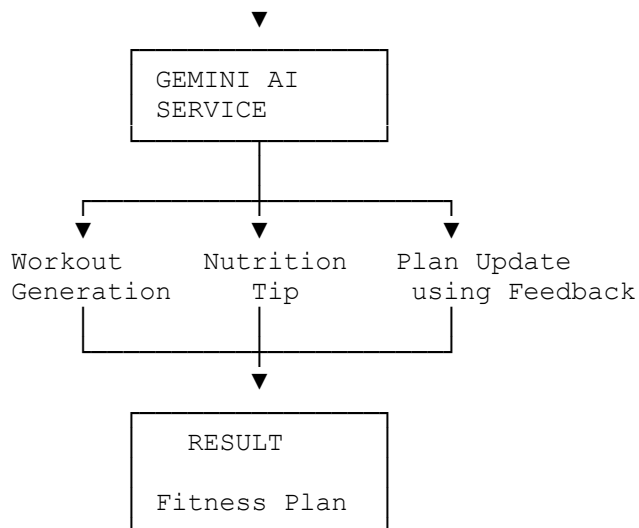
This function is used when the user provides feedback.

The system uses the feedback to modify the workout plan and generate an updated version.

11. AI Integration Flow

Figure 4: Generative AI Integration





This design separates AI-related operations from the main application logic.

12. Backend Route Design

FastAPI routes are designed to manage different user actions.

The home route provides the main user interface where the user can enter fitness information.

/generate-workout

This route processes the user's fitness information and initiates workout-plan generation.

/submit-feedback

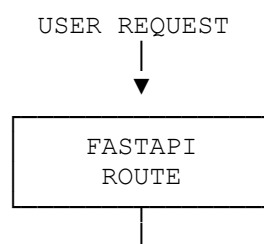
This route receives user feedback and initiates the workout-plan update process.

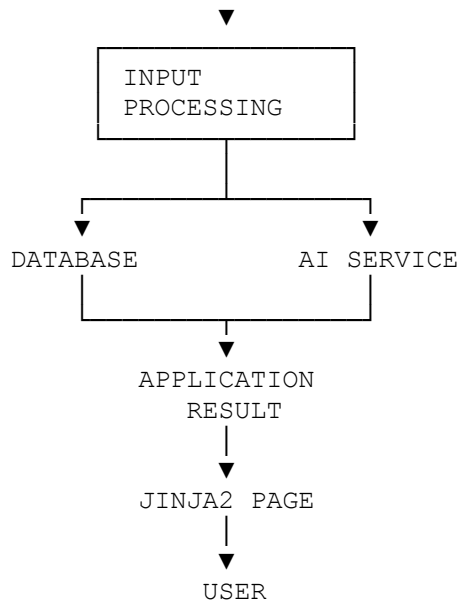
/view-all-users

This route provides access to stored users and plan information as documented in the source project.

13. Backend Processing Flow

Figure 5: FastAPI Request Processing





14. Frontend Design

The FitBuddy frontend is designed around a fitness-oriented interface.

The source project describes a UI using:

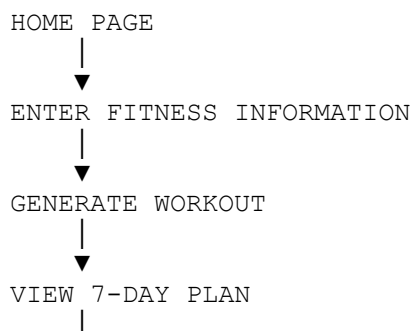
- Dark background
- Roboto font
- Flexbox layout
- Rounded corners
- Shadows
- Hover effects
- Responsive media queries

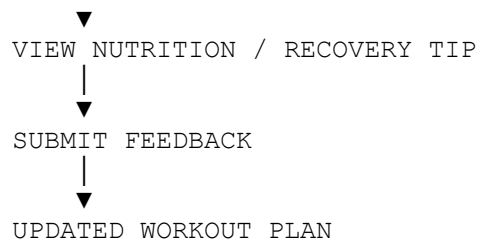
The purpose of the design is to provide a visually organized fitness application interface.

15. User Interface Structure

The frontend is designed to support the major user activities.

User Flow

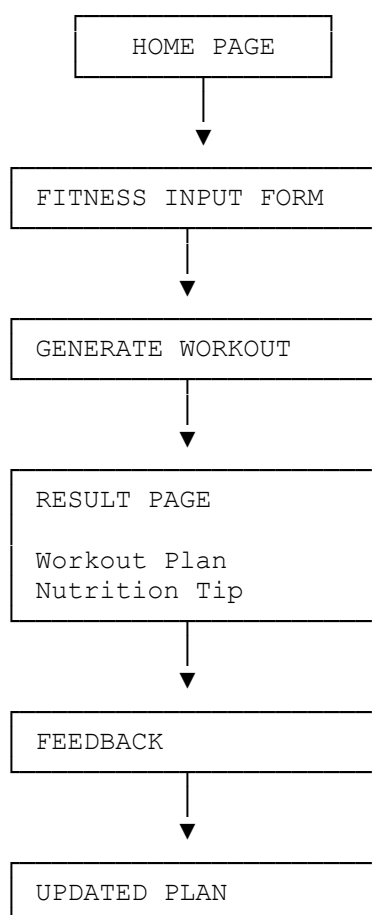




This structure creates a simple sequence from input to personalized output and feedback.

16. Frontend Navigation Design

Figure 6: FitBuddy User Interface Flow



17. Workout Plan Design

The generated workout plan is designed around the user's fitness information.

The system is intended to generate a **7-day workout plan**.

The plan generation considers information such as:

- User age
- User weight
- Fitness goal
- Workout intensity

The resulting plan is displayed through the frontend.

The documented fitness goals include:

- Weight loss
- Muscle gain
- General wellness

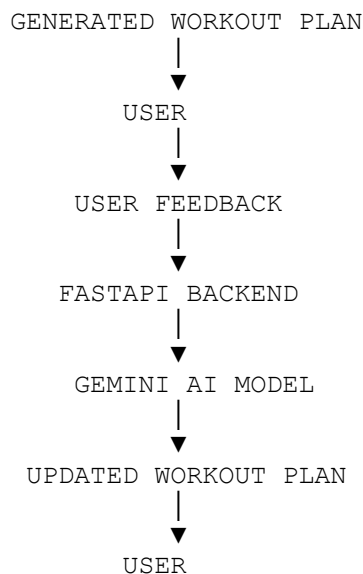
18. Feedback and Plan Revision Design

Feedback is an important part of the FitBuddy design.

After receiving the generated workout plan, the user can provide feedback.

The feedback is sent to the backend and passed to the plan-update functionality.

Figure 7: Feedback Loop

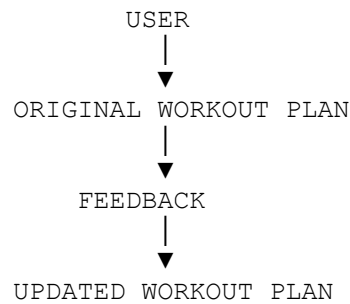


This design creates an iterative process in which the generated plan can be modified according to user feedback.

19. Original and Updated Plan Design

The source project specifically includes functionality for maintaining the original and updated workout plans.

The conceptual design is:



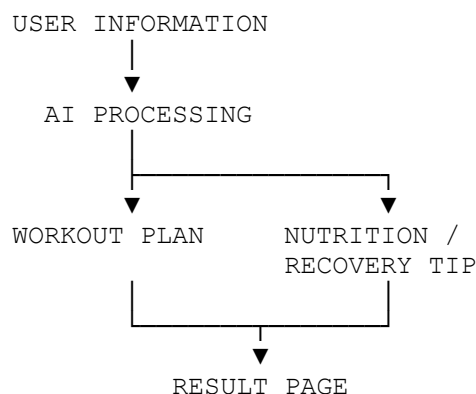
Keeping the original and updated versions makes it possible to distinguish between the first generated plan and the plan produced after feedback.

20. Nutrition and Recovery Design

FitBuddy also includes a nutrition/recovery tip as part of the generated result.

The nutrition functionality is separated from the main workout-generation functionality.

Design Flow



The documented source describes the nutrition output as a concise tip.

21. Configuration Design

The project design includes configuration for connecting the application to the Generative AI service.

The documented project uses an environment variable for the Google API key:

GOOGLE_API_KEY

The API key is kept outside the main application source code through environment configuration.

This design allows the application to access the AI service without directly placing the key in the application logic.

22. Deployment Design

The documented project is designed to run locally using **Uvicorn**.

The local application is accessed through:

`http://127.0.0.1:8000`

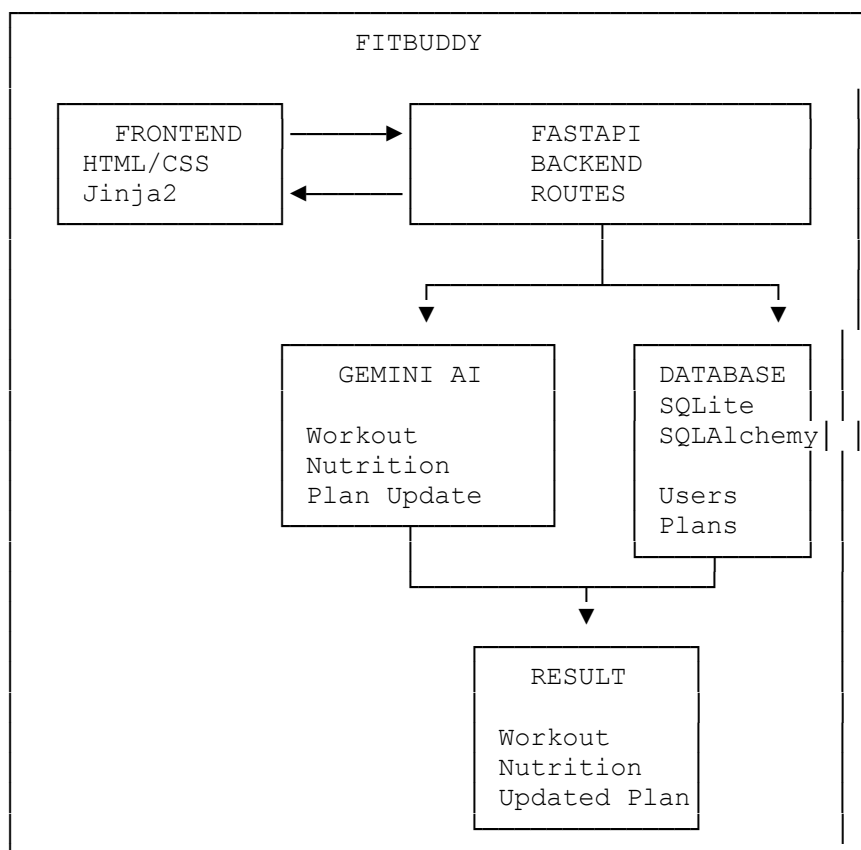
FastAPI also provides an API documentation interface at:

`/docs`

The deployment design therefore supports local development, testing, and demonstration.

23. Design Interaction Between Components

Figure 8: Complete FitBuddy Component Interaction



24. Design Decisions

The following design decisions are derived from the documented FitBuddy project:

24.1 FastAPI

FastAPI is used as the backend framework because the project requires routing and processing of user input.

24.2 Gemini

Google Gemini is selected for the Generative AI functionality used to create and revise fitness-related content.

24.3 SQLite

SQLite is used for persistent local storage of user and workout-plan information.

24.4 SQLAlchemy

SQLAlchemy is used as the database interaction layer.

24.5 Jinja2

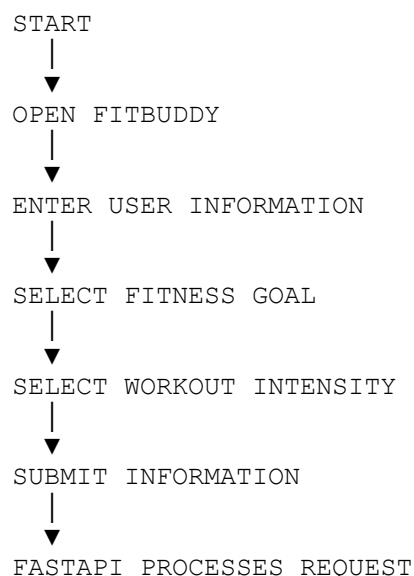
Jinja2 templates are used to generate dynamic frontend pages from backend data.

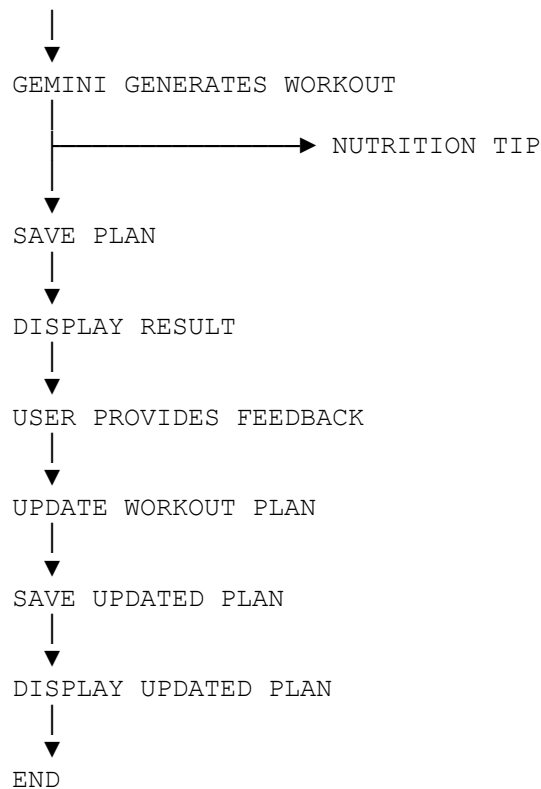
24.6 Modular AI Functions

The AI functionality is separated into different functions for workout generation, nutrition tips, and workout-plan updates.

25. Design of the Complete User Journey

The complete system design can be summarized as follows:





This workflow connects the major design elements into one complete application process.

26. Phase 3 Design Deliverables

At the end of the Project Design Phase, the following design elements are defined:

1. High-level system architecture.
2. Application component structure.
3. Frontend and backend interaction.
4. AI integration design.
5. Database design.
6. User input and output flow.
7. FastAPI route design.
8. Workout-generation flow.
9. Nutrition/recovery flow.
10. Feedback and plan-update flow.
11. Original and updated plan storage concept.
12. User interface navigation.
13. Configuration concept.
14. Local deployment concept.

These design elements provide the technical foundation for the development phase.

27. Transition to Phase 4 – Project Planning

After completing the Project Design Phase, the next stage is **Phase 4 – Project Planning**.

The design defines **how the system should be structured**, while the planning phase defines **how the project will be developed and organized**.

Project planning will include:

- Task identification
- Development sequence
- Technology planning
- Resource planning
- Time planning
- Milestone planning
- Risk identification
- Testing planning
- Documentation planning

The planning phase will convert the FitBuddy design into an organized development roadmap.

28. Phase 3 Conclusion

The Project Design Phase establishes the technical structure of the FitBuddy – AI Fitness Plan Generator.

The design connects the user interface, FastAPI backend, Generative AI service, database, and feedback mechanism into a single system. The architecture supports personalized workout generation, nutrition/recovery guidance, user feedback, and workout-plan revision.

The design also establishes the interaction between frontend pages, backend routes, AI functions, and database operations.

Therefore, Phase 3 provides the necessary technical blueprint for moving from requirements into actual implementation during the **Project Development Phase**.

